



Detection and Analysis of Silent Decision Degradation in AI-Assisted Inventory Systems

A Preliminary Project Report

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Project Overview & Motivation

- **Motivation**

- AI-based decision systems are widely used in inventory, logistics, and operations
- Evaluation is usually based on:
 - Accuracy
 - Cost
 - Average service level
- These metrics assume:
 - Stable environments
 - Immediate failure visibility

Key Question

How can we identify and analyze gradual, silent degradation in AI-assisted decision systems before overt failure occurs?

- **Core Concern**

- In real operational settings:
 - Demand evolves gradually
 - Signals are delayed or incomplete
 - Structural dependencies exist
- Under such conditions:
 - Systems may continue to operate
 - Decisions appear plausible
 - **Decision quality degrades silently**



Problem Statement & Objectives

- **Problem Statement**

- Existing research focuses on:
 - Prediction accuracy degradation
 - Cost or service-level optimization
- There is limited focus on:
 - **Decision behavior over time**
 - **Gradual degradation without metric collapse**
- Silent degradation is dangerous because:
 - It is hard to detect
 - It accumulates over time
 - It leads to delayed operational failure

Project Objectives

Model a realistic inventory decision environment

Build a transparent baseline decision system

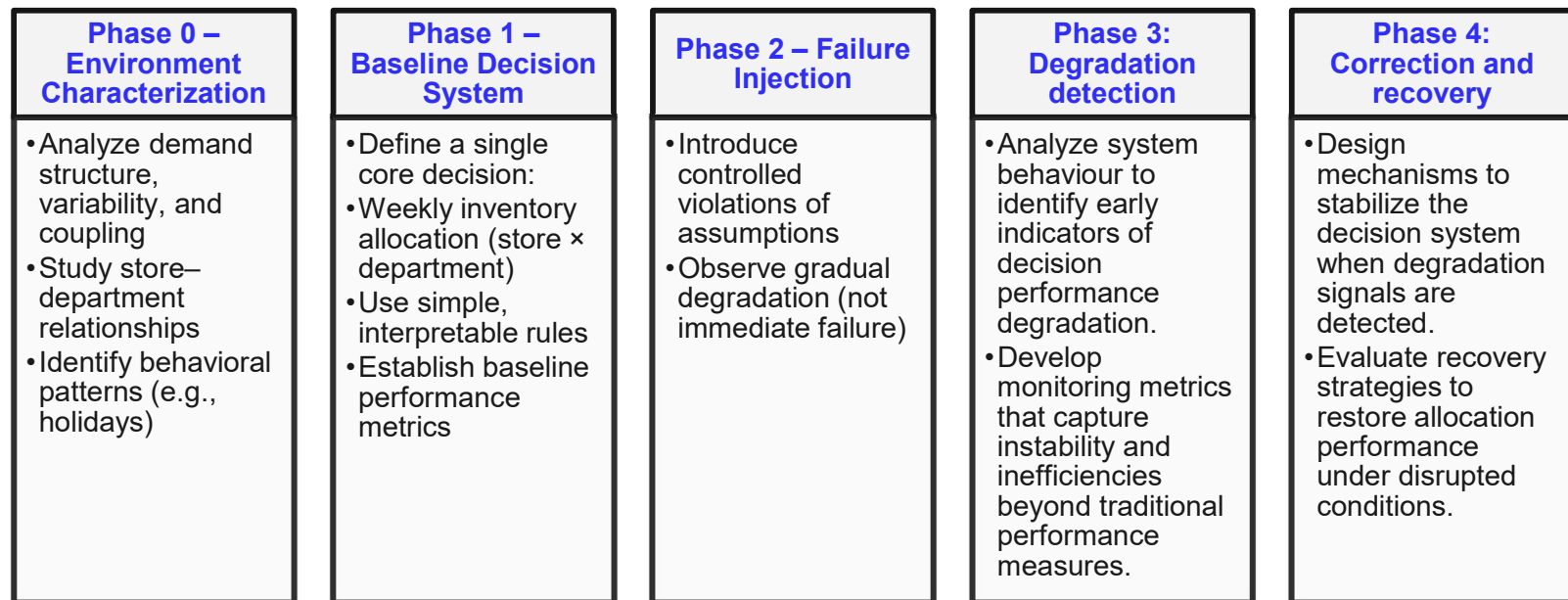
Systematically inject controlled failures

Observe how decision behavior degrades

Prepare a framework for degradation detection and correction



Methodology (Overall Structure)



*Inventory is used as a **testbed**, not the final objective.*



System Setup & Baseline Model

Phase 0

- Walmart Store Sales dataset used for modeling inventory demand
- Data preprocessing, exploratory analysis, and demand variability assessment
- Identification of store-department demand patterns and seasonal effects

Phase 1

- Developed baseline inventory allocation decision system
- Defined performance metrics: service level, unmet demand, overstock
- Baseline system establishes reference behaviour under stable conditions

Results

- Baseline service level ≈ 0.92
- Demand heterogeneity observed across stores
- Holiday periods cause strong demand volatility



Decision Health Panel		
Fill Rate (Weighted)	Average Service Level	Forecast MAPE
0.930	0.934	0.070
Demand-weighted fraction of demand satisfied. More honest than averaging service level.		
Mean Allocation Volatility	Allocation CV	Overstock Ratio
28,889.26	0.073	0.065
Average week-to-week change in allocation. High values imply unstable decision policy.		
Capital Efficiency	Cumulative Economic Loss	
1.006	3,555,674.77	
Sales-to-allocation ratio. Values below 1 indicate systemic oversupply.		
Total Sales	Average Weekly Loss	Loss Ratio
222,402,808.85	24,864.86	0.016
Aggregate realized demand across selected filters.		
Mean weekly economic damage over selected period.		
Proportion of total sales lost due to unmet demand and overstock.		



Failure Experiments & Experimental Dashboard

Phase 2 work

Introduced controlled disruptions to simulate real operational conditions

Implemented failure scenarios such as demand drift and delayed information

Evaluated how allocation decisions change under disturbed environments

Observations

Allocation behaviour becomes unstable under disrupted conditions

Inefficiencies appear despite stable aggregate metrics

Indicates possibility of **silent degradation in decision performance**

Dashboard

Developed an interactive **Streamlit** experimental dashboard

Enables visualization of allocation performance and degradation behaviour

Used to monitor metrics, compare scenarios, and analyze system response



Metric Comparison

	Metric	Baseline Model	After Failure	Δ Change
0	total_sales	222402808.850000	215992168.800000	6410640.050000
1	total_allocated	221089873.526200	216657007.326600	4432866.199600
2	total_loss	3555674.769500	3123455.687400	432219.082100
3	loss_ratio	0.016000	0.014500	<0.001000
4	avg_weekly_loss	24864.858500	22470.904200	2393.954300
5	mean_volatility	28889.262200	28585.351600	303.910600
6	avg_service	0.934100	0.941900	<0.001000
7	mape	0.070100	0.072000	0.001900
8	fill_rate	0.929700	0.937500	0.007800
9	overstock_ratio	0.064800	0.065300	0.000500



Future Plan, Timeline & Significance

- **Future Plan**

- Phase 3:
 - Design degradation detection signals
 - Compare against traditional metrics
- Phase 4:
 - Explore correction and recovery mechanisms
- Extend analysis across decision variants

- **Significance**

- Shifts focus from *model accuracy* to *decision integrity*
- Addresses a realistic, under-studied failure mode
- Provides a reusable experimental framework
- Applicable beyond inventory systems

Goal: Detect decision degradation before performance collapse.

